Application Flow Document v2.0

CloudWise AI

Version: 2.0
Status: Active Development
Last Updated: June 2026

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1. Purpose

This document defines the complete application flow of CloudWise AI, including user interactions, frontend behavior, backend processing, database operations, AWS integrations, LocalStack integrations, recommendation generation, dashboard analytics, and future AI workflows.

The goal is to provide a clear understanding of how data moves through the system.

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2. High-Level System Flow

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3. User Authentication Flow

Registration Flow

```
User
 ↓
Signup Form
 ↓
Frontend Validation
 ↓
POST /auth/register
 ↓
Backend Validation
 ↓
Hash Password
 ↓
Store User
 ↓
Success Response
 ↓
Login Page
```

Login Flow

```
User
 ↓
Login Form
 ↓
POST /auth/login
 ↓
Verify Credentials
 ↓
Generate JWT
 ↓
Return Token
 ↓
Store Token
 ↓
Redirect Dashboard
```

Forgot Password Flow

```
User
 ↓
Forgot Password
 ↓
Enter Email
 ↓
POST /auth/forgot-password
 ↓
Generate Reset Token
 ↓
Send Email
 ↓
User Opens Link
 ↓
Reset Password Form
 ↓
POST /auth/reset-password
 ↓
Update Password
 ↓
Login
```

---

4. Cloud Connection Flow

CloudWise supports two environments:

AWS Environment

```
Real AWS Account
```

LocalStack Environment

```
Local Development Cloud
```

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5. AWS Connection Flow

```
User
 ↓
Connect Cloud
 ↓
Enter AWS Credentials
 ↓
Submit
 ↓
POST /cloud/connect
 ↓
connect_aws()
 ↓
STS Validation
 ↓
Account Verification
 ↓
Save Cloud Account
 ↓
Success
```

---

6. LocalStack Connection Flow

```
User
 ↓
Enable LocalStack Toggle
 ↓
Frontend Sets

use_localstack=true

 ↓
POST /cloud/connect
 ↓
connect_aws()
 ↓
endpoint_url=http://localhost:4566
 ↓
LocalStack STS Validation
 ↓
Account Verification
```

```
 ↓
Success
```

---

7. Resource Discovery Flow

After cloud connection:

```
User
 ↓
Click Scan Resources
 ↓
POST /cloud/scan
 ↓
Cloud Scan Service
```

EC2 Discovery

```
discover_ec2()
 ↓
Describe Instances
 ↓
Normalize Data
 ↓
Create Resource Objects
```

Collected:

- Instance ID
- Instance Type
- Status
- Region
- Tags

EBS Discovery

```
discover_ebs()
 ↓
Describe Volumes
```

```
 ↓
Normalize Data
 ↓
Create Resource Objects
```

Collected:

- Volume ID
- Size
- Attachments
- Encryption
- Status

---

8. Resource Persistence Flow

After discovery:

```
Discovered Resources
 ↓
Normalization
 ↓
ResourceInventory Mapping
 ↓
Database Insert
 ↓
resource_inventory
```

Stored Fields:

```
resource_id
resource_name
resource_type
region
status
tags
metadata_json
```

---

9. Metrics Collection Flow

```
Resource IDs
 ↓
CloudWatch Request
 ↓
CPU Utilization
 ↓
Metrics Processing
 ↓
metrics Table
```

Current Metric:

```
CPU Utilization
```

Future:

```
Memory
Disk
Network
```

---

10. Cost Analysis Flow

```
Cloud Account
 ↓
Cost Explorer
 ↓
Daily Costs
 ↓
Monthly Costs
 ↓
Cost Aggregation
 ↓
Dashboard Metrics
```

Generated Data:

- Current Spend

- Monthly Spend
- Cost Trends

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11. Recommendation Engine Flow

Input Sources

```
Resources
+
Metrics
+
Costs
```

↓

Recommendation Engine

Rule 1

Idle EC2 Detection

```
CPU < 5%
```

↓

Recommendation

```
Shutdown Instance
```

Rule 2

Underutilized EC2

```
CPU < 20%
```

↓

Recommendation

```
Resize Instance
```

Rule 3

Unattached EBS

```
No Active Attachments
```

↓

Recommendation

```
Delete Volume
```

---

12. Recommendation Storage Flow

```
Recommendation Engine
 ↓
Generate Recommendation
 ↓
Calculate Savings
 ↓
Store Recommendation
 ↓
recommendations Table
```

Stored Data:

```
priority
estimated_savings
description
recommendation_type
```

---

13. Dashboard Flow

Dashboard Load

```
User
 ↓
Dashboard Page
 ↓
GET /dashboard
 ↓
Backend Aggregation
```

Backend Reads:

```
resource_inventory
recommendations
metrics
cost_data
```

Backend Calculates:

```
Total Resources
Monthly Spend
Potential Savings
Health Score
```

↓

Response

↓

Frontend Dashboard

---

14. Dashboard Components Flow

KPI Cards

```
Dashboard API
 ↓
Total Resources

Dashboard API
 ↓
Monthly Spend

Dashboard API
 ↓
Potential Savings

Dashboard API
 ↓
Health Score
```

Cost Trend Chart

```
Cost Data
 ↓
Aggregation
 ↓
30 Day Trend
 ↓
Area Chart
```

Resource Distribution

```
Resource Inventory
 ↓
Group By Type
 ↓
Visualization
```

---

15. Resource Explorer Flow

```
Resources Page
 ↓
GET /cloud/resources
 ↓
Database Query
 ↓
Paginated Results
 ↓
Frontend Table
```

Features:

- Search
- Filter
- Sort

---

16. LocalStack Demo Flow

Purpose:

Prevent empty dashboards during development and demonstrations.

Flow

```
Start LocalStack
 ↓
Create Mock Resources
 ↓
Run Discovery
 ↓
Store Resources
 ↓
Generate Recommendations
 ↓
Populate Dashboard
```

Mock Resources:

- EC2
- EBS

---

17. Future Forecasting Flow

```
Historical Costs
 ↓
Forecast Engine
 ↓
Prediction Model
 ↓
7 Day Forecast

30 Day Forecast

90 Day Forecast
```

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18. Future AI Copilot Flow

```
User Question
 ↓
Copilot Interface
 ↓
Context Builder
```

Context Sources:

- Resources
- Metrics
- Costs
- Recommendations
- Forecasts

↓

LLM Processing

↓

AI Response

Example:

```
Why is my AWS bill increasing?
```

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19. Report Generation Flow

```
User
 ↓
Generate Report
 ↓
Collect Dashboard Data
 ↓
Collect Recommendations
 ↓
Generate Summary
 ↓
Create PDF
 ↓
Download Report
```

---

20. Error Handling Flow

AWS Failure

```
AWS Error
 ↓
Capture Exception
 ↓
Return User Friendly Message
```

LocalStack Failure

```
Connection Error
 ↓
Show Diagnostic Message
```

Database Failure

```
Database Error
 ↓
Log Error
 ↓
Return Safe Response
```

---

21. Complete End-to-End User Flow

```
Signup
 ↓
Login
 ↓
Connect AWS / LocalStack
 ↓
Validate Credentials
 ↓
Scan Resources
 ↓
Discover EC2 & EBS
 ↓
Store Resources
 ↓
Collect Metrics
 ↓
Analyze Costs
 ↓
Generate Recommendations
 ↓
Populate Dashboard
 ↓
View Insights
```

```
 ↓
Forecast Costs
 ↓
Generate Reports
 ↓
Optimize Infrastructure
```

Application Status

Current Production Features

✅ Authentication
✅ Forgot Password
✅ AWS Integration
✅ LocalStack Integration
✅ EC2 Discovery
✅ EBS Discovery
✅ Resource Inventory
✅ Dashboard Analytics
✅ Recommendation Engine

Planned Features

🔄 Forecasting Engine
🔄 AI Copilot
🔄 PDF Reports
🔄 S3 Discovery
🔄 RDS Discovery

Document Status:

Approved

Version:

2.0